Steadfast: Zero Unaccounted Collections Calls

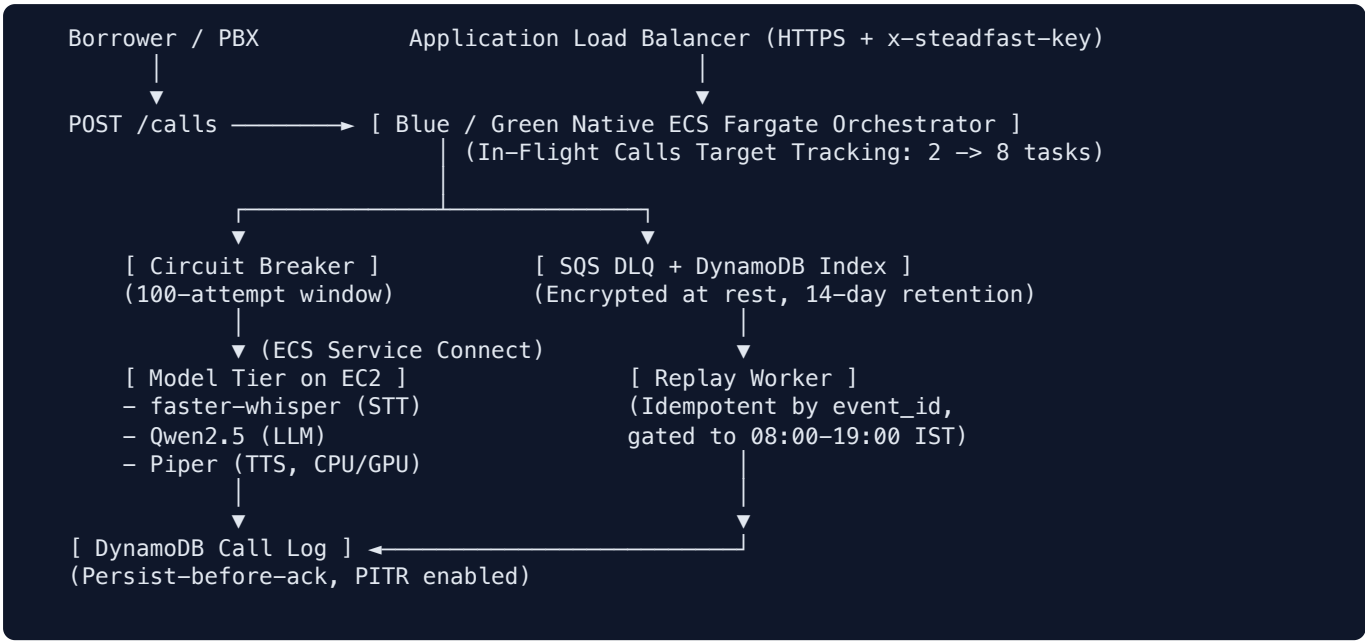
Predixion AI × TalentX Open-Weight Collections Agent Challenge (AWS FDE Track)

Architecture, Resilience Engineering & Production Deployment on AWS ap-south-1 | **Repository:** https://github.com/maannaan/steadfast (Apache-2.0)

EXECUTIVE SUMMARY

In digital lending operations governed by Reserve Bank of India (RBI) regulations, an unlogged collections call is an audit failure and a compliance violation. Yet conversational voice pipelines chaining Speech-to-Text (STT), Large Language Models (LLM), and Text-to-Speech (TTS) inherently face unpredictable latency, high concurrency, and downstream failure rates of 20% or more. **Steadfast** is a carrier-grade inference orchestration service built to enforce a singular guarantee: **zero unaccounted calls**. Every borrower interaction ends deterministically in the **call log** (with full telemetry and business outcome) or in the **dead-letter queue (DLQ)** (with structured error reasons, ready for automated replay). No call is ever silently dropped.

SYSTEM ARCHITECTURE (AWS AP-SOUTH-1)



- **Persist-Before-Ack & Idempotency:** Calls are keyed by deterministic `event_id`. HTTP 200 is issued only after state persistence (DynamoDB on AWS, SQLite with `synchronous=FULL` locally). Downstream failures return HTTP 200 with recorded DLQ status; HTTP 500 is reserved exclusively for unrecordable errors so clients know to resend.
- **Full-Jitter Exponential Backoff:** 3 attempts with full jitter (0.2 s base, 2.0 s cap), bounded by a 2.5 s attempt timeout and an 8.0 s call deadline. A rolling 100-attempt per-task circuit breaker trips at 50% failure, failing fast into the DLQ.
- **Target Tracking on In-Flight Calls:** Scaled on `InFlightCalls` per task (target: 20 calls/task) via Little's Law, published using CloudWatch Embedded Metric Format (EMF) logs without `PutMetricData` API charges.
- **Zero-Downtime Blue/Green & Multi-AZ:** Production traffic shifts over a 2-minute bake period with header-based pre-verification (`X-Deploy-Test: 1`) and automatic alarm rollbacks. Tasks are balanced across Availability Zones with clean draining.

- **RBI & DPDP Act Compliance:** 100% data residency in Mumbai (`ap-south-1`), zero PII in log streams, and strict campaign calling window gating (`08:00–19:00 IST`) for both real-time traffic and DLQ replay workers.

EMPIRICAL VALIDATION & TEST RESULTS

Test Scenario	Volume / Speed	Injected Conditions	Outcome & Scalability	Unaccounted
Local Independent	6,083 calls (60×)	20% random downstream 503s	4,876 1st-try, 1,163 retried, 44 DLQ	**0**
Local Outage Burst	6,083 calls (60×)	15s outage every 60s (4 trips)	3,161 1st-try, 725 retried, 2,197 DLQ	**0**
AWS Campaign Window	6,083 calls (6×)	6-turn calls, 20% faults, autoscaling	Tasks scaled 2 → 8 → 2; DLQ drained clean	**0**
AWS Multi-AZ Drill	Live campaign	Terminated all tasks in AZ-a	AZ-b absorbed load, 0 client errors	**0**
AWS Blue/Green Shift	Live campaign	Good shift vs. bad tag rollback	Green tested; broken tag rolled back auto	**0**
Open-Weight CPU Tier	1,037 calls (1×)	4× overload on Whisper+Qwen+Piper	Fast 429 shedding directly to DLQ	**0**

COST ECONOMICS & INFRASTRUCTURE EFFICIENCY

- **Measured Session Cost: $2.70 total** on AWS `ap-south-1` via Cost Explorer receipts (Fargate: $0.88; NAT Gateway: $0.86; EC2 model host: $0.47; ALB: $0.31; VPC/IP: $0.16; DynamoDB/SQS: $0.02; CloudWatch EMF: < $0.01).
- **Budget Cap Compliance:** Spent less than 10% of the ₹2,500 (~$30.00) challenge reimbursement limit.
- **Unit Cost: ~$0.000074 per call** (~6.2 paise per 100 calls) during peak autoscaling load.
- **Programmatic Teardown Verification:** `scripts/teardown.sh` queries AWS APIs directly for all tagged resources at session completion: **RESULT: PASS**.